

Algebra 1
Unit 9
Lesson 4 Practice

Name Answer key
Date _____
Period _____

Organizers of a music festival charge \$60 for admission. The organizers polled the attendees to determine how increasing the price would affect attendance, and therefore affect profit. They discover that if they reduce the price by \$1, their profit would be maximized at \$600,000. However, if they raised the price by \$1, they would make \$560,000 in profit. The relationship between the music festival's profit, y , and the difference in the admission price, x , can be modeled by a quadratic equation.

1. Identify the vertex and another point from the context.

a. Vertex: (59, 600,000)

b. Point: (61, 560,000)

2. Find the value of a for the quadratic equation modeling the music festival's profit.

$$\begin{aligned} y &= a(x - 59)^2 + 600,000 \\ 560,000 &= a(61 - 59)^2 + 600,000 \\ 560,000 &= 4a + 600,000 \\ \frac{-600,000}{4} &= \frac{4a}{4} \end{aligned}$$

$$a = -10,000$$

3. Write the quadratic equation that models the music festival's profit in vertex form.

$$y = -10,000(x - 59)^2 + 600,000$$

4. Rewrite the equation in standard form.

$$\begin{aligned} y &= -10,000(x - 59)(x - 59) + 600,000 \\ &= -10,000(x^2 - 118x + 3481) + 600,000 \\ &= -10,000x^2 + 1,180,000x - 34,810,000 + 600,000 \\ y &= -10,000x^2 + 1,180,000x - 34,210,000 \end{aligned}$$

A statistician for a running shoe company analyzes data from the years 2007 to 2014 to help inform the company about trends. The relationship between the number of people ages 6 to 17, in millions, y , who listed running as a hobby for x years since 2007 can be modeled by a quadratic equation. In 2011, 4 years into the study, the maximum number of runners in this age group was recorded at 10,000,000. Just one year earlier, there were 9,860,000 runners recorded in this age group. **10 million**

5. Identify the vertex and another point from the context.

a. Vertex: (4, 10)

b. Point: (3, 9.86)

6. Find the value of a for the quadratic equation that models the relationship.

$$\begin{aligned} y &= a(x - 4)^2 + 10 \\ 9.86 &= a(3 - 4)^2 + 10 \\ 9.86 &= a(-1)^2 + 10 \\ 9.86 &= a + 10 \\ \underline{-10} \quad \underline{-10} \\ a &= -0.14 \end{aligned}$$

7. Write the quadratic equation that models the relationship in vertex form.

$$y = -0.14(x - 4)^2 + 10$$

A quadratic equation can model the height above the ground, y , in feet, of a toy rocket after x seconds that it is launched from the top of a building. The rocket reaches a maximum height of 256 feet above the ground after 2 seconds of flight time. The rocket hits the ground 6 seconds after its launch.

8. Identify the vertex and another point from the context.

a. Vertex: (2, 256)

b. Point: (6, 0)

9. Find the value of a for the quadratic equation modeling the rocket's height.

$$\begin{aligned} y &= a(x - 2)^2 + 256 \\ 0 &= a(6 - 2)^2 + 256 \\ 0 &= 16a + 256 \\ \underline{-256} \quad \underline{-256} \\ -256 &= 16a \\ \underline{16} \quad \underline{16} \\ a &= -16 \end{aligned}$$

10. Write the quadratic equation that models the rocket's height in vertex form.

$$y = -16(x - 2)^2 + 256$$